Sara Jiang, DATA 11800/1, Final Project

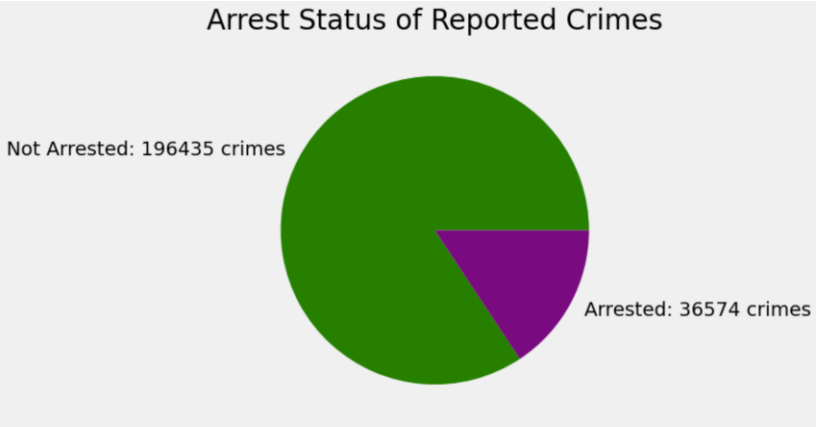
Introduction of Dataset:

The dataset I am analyzing contains crime reports in Chicago, excluding murders, from one year before February 16, 2026 to one week before February 16, 2026. The data is extracted from the Citizen Law Enforcement Analysis and Reporting (CLEAR) system. The CLEAR system exists to better manage police and improve connections between the Chicago Police Department and other criminal justice agencies, and CPD and the communities they serve. (Skogan et al., 2004) This data came from a retroactive study. There are 17 variables in the dataset, such as whether someone was arrested, the approximate longitude, and the approximate latitude of a reported crime. There are 233,009 observations. This data samples every reported crime within the time frame, and it is biased for many reasons, as it is well known that police target certain socioeconomic groups more than others, and a certain type of crime (murder) is explicitly excluded.

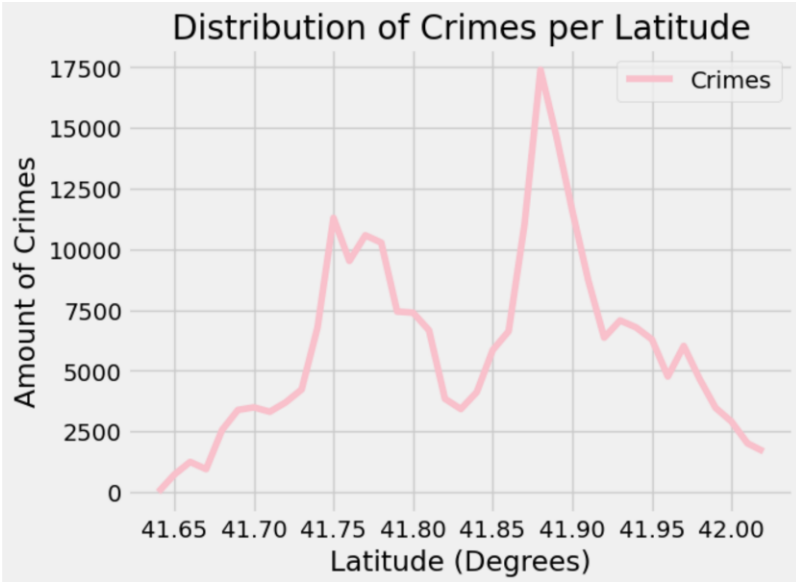
Characteristics of Sample:

The crimes in this dataset have many variables associated with them, though the ones I will focus on in this project are the following:

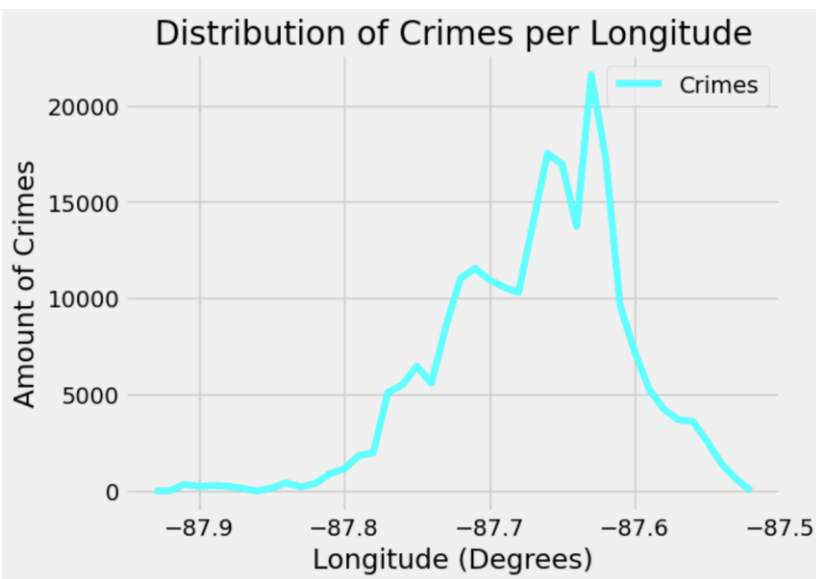
- Arrests
  - Originally a Categorical Variable
    - Changed to a Numerical Variable for ease of analysis
  - Whether someone was arrested for a reported crime
    - 0 means not arrested, 1 means arrested
  - Integer
  - Skewed Left



- Latitude
  - Numerical Variable
    - Originally a Continuous Variable
    - I changed this to a Discrete Variable by rounding for ease of analysis
  - The latitude of the crime
  - Float
  - Bimodal



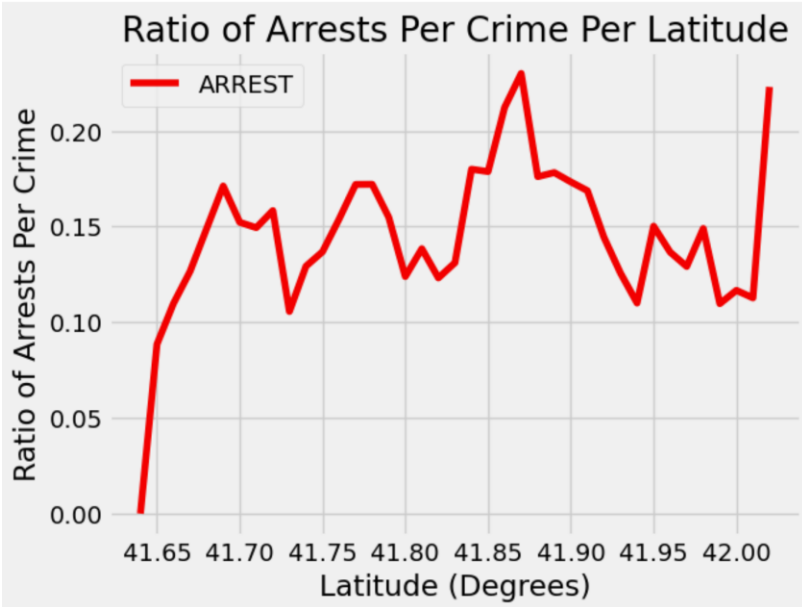
- Longitude
  - Numerical Variable
    - Originally a Continuous Variable
    - I changed this to a Discrete Variable by rounding for ease of analysis
  - The longitude of the crime
  - Float
  - Skewed Left



Relationships between Variables:

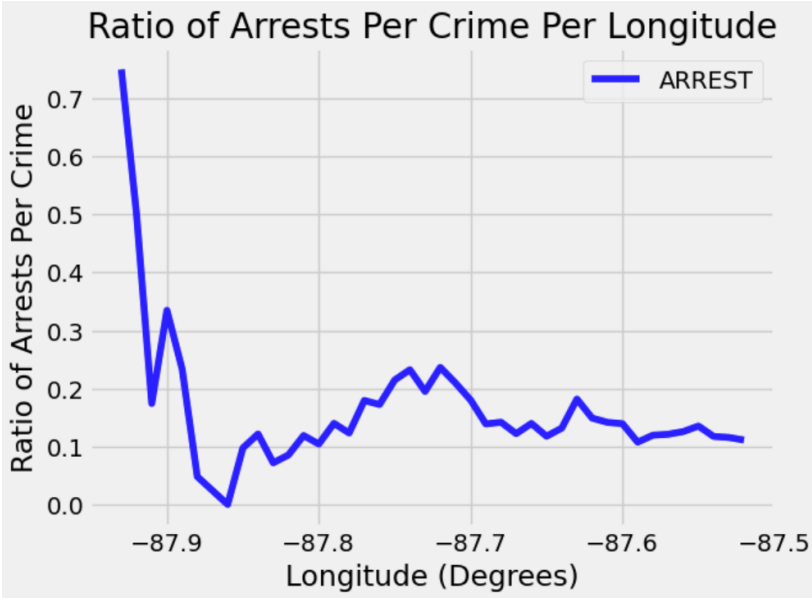
Arrests vs. Latitude:

With grouping, I created a column that has the ratio of arrests to total reported crimes per latitude. The right peak from latitude alone (see Distribution of Crimes per Latitude) is visible in the association between arrests and latitude, but the left peak has disappeared. This new distribution is skewed left.



Arrests vs. Longitude

With grouping, I created a column that has the ratio of arrests to total reported crimes per longitude. This distribution looks very similar to longitude alone (see Distribution of Crimes per Longitude) This new distribution is skewed right.



Hypothesis Test:

The relationships I've discovered show that someone is less likely to be arrested for a crime they commit in the south half of the city than in the north half of the city. This is because the data for arrests versus latitudes is skew right. Additionally, the left/south end of the data juts down and the right/north end of the data juts up.

Null Hypothesis: There is no difference between the probability of getting arrested in the south half of the city and getting arrested in the north half of the city, given that you commit a crime that is reported within one year to one week before February 16, 2026.

Alternate Hypothesis: There is a negative difference between the probability of getting arrested in the south half of the city and getting arrested in the north half of the city, given that you commit a crime that is reported within one year to one week before February 16, 2026.

Test Statistics:

- Stimulated probability of getting arrested in the north half of the city, given that you commit a crime that is reported within one year to one week before February 16, 2026
- Stimulated probability of getting arrested in the south half of the city, given that you commit a crime that is reported within one year to one week before February 16, 2026.

Conclusion:

In this project, I observed the arrest status, longitude, and latitude of reported crime in Chicago. The distribution of latitudes from crime in Chicago is bimodal. The distribution of arrest from crime in Chicago, the distribution of longitudes from crime in Chicago and the distribution of the ratio between arrests and crimes and latitude in Chicago is skewed left. The ratio between the distribution of the ratio between arrests and crimes and longitude in Chicago is skewed right. I am wondering how many crimes go unreported and if locations affects that, because such crimes don't appear in the data.

Bibliography:

Research & Development Divison. Retrieved February 16, 2026, from "https://data.cityofchicago.org/Public-Safety/Crimes-One-year-prior-to-present/x2n5-8w5q/about_data."

Skogan, W., Hartnett, S., Dubois, J., Bennis, J., Kim, S. Y., Rosenbaum, D., Graziano, L., & Stephens, C. (n.d.). *Policing Smarter Through IT: Learning From Chicago's Citizen and Law Enforcement Analysis and Reporting (CLEAR) System*. https://portal.cops.usdoj.gov/ResourceCenter/content.ashx/cops-w0152-pub.pdf